Anmol Singh Yadav

AI Platform Security Engineer | Security Architecture

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SUMMARY

Security-focused Platform Engineer specializing in **cloud-native security, AI security, vulnerability research, and security architecture**. Experienced in securing production Kubernetes and container platforms and designing security controls for **enterprise AI platforms and agentic systems** across cloud and on-premises environments. Hands-on across **Linux, containers, eBPF, GCP, application security, OSS supply-chain security, and LLM security**, with a focus on translating attack paths into scalable security controls.

CAREER

ISS STOXX | Technology Innovation Lab *Jan 2023 – Present*
Platform Security Engineer *Mumbai, India*

- Designed and led adoption of a hardened container runtime architecture, eliminating a validated container escape path across **160 production Kubernetes clusters** and **53 Linux hosts** without operational disruption.
- Architected security controls for a **2,000-user Enterprise AI Platform**, securing interactions between GCP-hosted AI agents and on-premises enterprise data through identity, API, network, data and execution controls.
- Authored the platform's **AI security architecture and threat model**, addressing agent identity, least privilege, tool governance, prompt injection, RAG isolation, data exfiltration, sandboxing, egress control and semantic guardrails.
- Implemented production controls using **GCP Agent Platform, Model Armor, Model Garden, Workload Identity Federation, Workforce Identity Federation, Apigee and Cloud Armor** to establish trust boundaries between users, agents, APIs and enterprise data.
- Architected an **SBOM-driven vulnerability intelligence system** across **1,000+ repositories**, reducing **9,200+ findings to fewer than 350 actionable risks** through contextual validation and LLM-assisted prioritization.
- Used the **Threat Validation Engine** for vulnerability assessments and successfully identified all projects affected by newly disclosed vulnerabilities in open-source packages, enabling teams to remediate vulnerable dependencies.
- Engineered an **eBPF-based runtime detection system** across **50+ production hosts**, providing kernel-level visibility into privilege escalation, OOM events, network activity and sensitive file access.
- Designed a **Kubernetes security assessment framework** for identifying privileged workloads, unsafe capabilities, default root execution, vulnerable packages and container escape paths; applied **Kata Containers and runcvm** for runtime isolation.

PROJECTS

- **SAST-AI-Harness**: Built an AI-assisted SAST framework as a **GitHub Copilot Skill**, orchestrating **10+ OSS security engines** with LLM-driven dataflow tracing, attacker-exposure analysis and iterative adversarial review for OWASP Top 10, memory-safety and novel vulnerability discovery across multi-language codebases.
- **pwnspectrum**: Built an automated cybersecurity news intelligence pipeline that scrapes **1,000+ security news articles**, applies ranking and relevance functions to identify the **top 20 cybersecurity events**, and publishes a curated security feed. [Feed]
- **Qwen3-4B Abliterated**: Researched **LLM refusal mechanisms and model abliteration**, producing and open-sourcing an experimentally modified Qwen model with **1,000+ downloads**, accompanied by technical research on refusal behaviour and guardrails. [Model] [Research]
- **kube-explor3r**: Developed a Kubernetes security assessment tool performing cluster, pod, Linux package and container escape assessments, with findings stored for continuous security monitoring.

TECHNICAL SKILLS

- **AI Security & Architecture**: AI Agents, LLM Security, Prompt Injection, Guardrails, Agent Isolation, Tool Governance, RAG Security, Threat Modeling, Secure Execution
- **Cloud & Platform Security**: GCP, Agent Platform, Model Armor, Model Garden, Apigee, Cloud Armor, Workload Identity Federation, Workforce Identity Federation, Kubernetes, Docker, containerd, runc, Kata Containers
- **Application & Supply Chain Security**: SAST, SCA, SBOM, Vulnerability Management, OSS Supply Chain Security, OWASP Top 10, Semgrep, CodeQL, OSV, Threat Validation
- **Linux & Runtime Security**: eBPF, Syscall Tracing, Kernel Security, Runtime Detection, Privilege Escalation, Container Escape Analysis, Network Security
- **Programming & Systems**: Go, Python, Bash, C/C++, Kafka, Redis, Grafana, OpenSearch

EDUCATION

Vellore Institute of Technology *2019 – 2023*
B.Tech in Computer Science Engineering, Specialization in Cyber Security & Digital Forensics **CGPA: 9.0 / 10**

SECURITY RESEARCH & SIGNALS

- **Conference Speaker**: Presented API Security at GRIMMCon, covering real-world vulnerabilities in API-driven systems.
- **Technical Writing**: Published **15+ security articles**, with individual articles reaching **30K+ readers**. [medium]
- **Offensive Security**: Ranked in the **Top 1% (0xD – God Level)** on TryHackMe through hands-on exploitation and CTF challenges. [tryhackme.com/p/IamLucif3r]